A Comprehensive Literature Survey on the Benchmark for Early-stage Parkinson's Disease Detection from Speech

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1 Introduction and Research Context

1.1 Background

Parkinson's Disease (PD) is the second most prevalent neurodegenerative disorder, affecting over 10 million people worldwide. One of the most promising avenues for non-invasive and scalable screening is the analysis of speech, as vocal impairment is among the earliest and most prevalent symptoms of the disease. Speech abnormalities may affect up to 90% of individuals with PD and can sometimes manifest years before prominent motor symptoms appear.

1.2 The Clinical Imperative for Early Detection

From a clinical perspective, detecting PD at an early stage is substantially more important than general binary classification (PD vs. Healthy Control), because:

- **Early intervention** may slow disease progression and improve patient management.
- Neurologists can already identify clinically manifested PD with high accuracy (91.5% within the first 5 years among experts).
- The true diagnostic gap lies in the **prodromal and early stages**, where symptoms are subtle.

As noted in contemporary reviews, speech alterations—including dysprosody, reduced pitch variability, and imprecise articulation—are among the earliest signs of PD and may serve as early markers for diagnosis and prediction.

2 The Problem: Fragmentation and Non-Comparability

2.1 Core Challenge

The literature reveals a critical gap: **speech-based EarlyPD detection remains comparatively underexplored and methodologically inconsistent**. While numerous studies claim to address "early detection," meaningful comparison across studies remains challenging due to substantial variation in datasets, speech tasks, evaluation protocols, and definitions of early-stage PD.

Table 1: Sources of variation across speech-based EarlyPD studies.

Dimension	**Variation Observed**
Datasets	Different languages, recording conditions, and sample sizes
Speech Tasks	Sustained vowels, DDK, sentence reading, spontaneous speech
Evaluation Protocols	Various cross-validation schemes and train/test splits
EarlyPD Definitions	MDS-UPDRS, H&Y scale, Time After Diagnosis (TAD), with no consistent rule

2.2 Inconsistent Definition of Early-Stage PD

Prior studies have employed different criteria to define "early-stage" PD:

1. **Hoehn & Yahr (H&Y) Scale-based:** Some studies use H&Y stage ≤ 2.
2. **MDS-UPDRS-based:** Used for measuring clinical outcomes but lacks widely accepted thresholds for staging.
3. **Time After Diagnosis (TAD)-based:** Cutoffs vary, for example ≤ 5 years or < 10 years.

Key Issue: The lack of a standardized definition confounds reported results and limits meaningful comparison of methodological contributions.

2.3 Inconsistent Validation Practices

A broader systematic review of 69 studies from 2020–2025 found that:

- Approximately 70% relied on internal validation, such as k-fold cross-validation or fixed train/test splits.
- Less than 15% conducted external validation.
- Subject-independent validation, including LOSO/LOOCV, was used in only 8 studies.
- Speaker-dependent data splitting remains common, potentially causing data leakage and overly optimistic performance estimates.

3 The Proposed Solution: A Standardized Benchmark

3.1 Benchmark Overview

To address the identified gap, Zhong et al. propose the **first benchmark for speech-based EarlyPD detection**, with the following key contributions:

1. **First benchmark** specifically targeting EarlyPD detection.
2. **Transparent, replicable protocol** with public resources.
3. **Multiple training conditions** to isolate the effects of training data composition.
4. **Multi-dimensional evaluation** for fine-grained comparisons.
5. **Open + Private tracks** balancing accessibility and real-world constraints.

3.2 Early-Stage PD Definition Adopted

The benchmark adopts a clear, clinically grounded definition from Burq et al. (2022).

Table 2: Definition of EarlyPD adopted by the benchmark.

Criterion	**Threshold**
Hoehn & Yahr Stage	≤ 2
Time After Diagnosis (TAD)	≤ 5 years

Rationale:

- **H&Y Scale:** Provides well-defined cutoff points with straightforward clinical interpretation.
- **MDS-UPDRS:** Rejected due to the lack of accepted staging thresholds.
- **Untreated PD (no Levodopa):** Rejected as clinically unrepresentative because it would bias the benchmark toward atypically mild disease.

3.3 Dataset Selection

3.3.1 Open Track (Researcher-Accessible)

The authors screened all potentially accessible PD speech datasets and found only two that met the benchmark criteria.

Table 3: Datasets selected for the Open Track.

Dataset	Speakers	PD	HC	Language	Key Metadata
PC-GITA	100	50	50	Spanish	MDS-UPDRS, H&Y, TAD, gender-balanced
NeuroVoz	108	53	55	Castilian Spanish	Rich clinical metadata

Excluded Dataset: EWA-DB was excluded because although its TAD inclusion criterion was less than 10 years, individual TAD values were not available.

3.3.2 Private Track

The private track uses **PERSPECTIVE-Base**, consisting of:

- 200 Dutch speakers with idiopathic PD.
- Pre-therapy recordings collected through a smartphone application.
- Sampling rate of 44.1 kHz and 16-bit resolution.
- Data obtained from a single-blind randomized controlled trial.

3.4 Speech Tasks

The benchmark covers three common speech tasks.

Table 4: Speech tasks included in the EarlyPD benchmark.

Task	Description	Clinical Relevance
Sustained Vowel	/a/ phonation	Isolates phonation features such as jitter, shimmer, and HNR
DDK	/pa-ta-ka/ repetition	Evaluates articulatory precision and motor speech speed
Sentence Reading	Standardized texts	Assesses prosody and fluency in connected speech

3.5 Training Data Settings

Four configurations isolate the effect of training data composition.

Table 5: Training data configurations.

Setting	Description	Purpose
AllPD	Full PD set including EarlyPD and non-EarlyPD	Baseline using all disease stages
AllPD-sub	Distribution-matched subset of AllPD	Match speaker count to EarlyPD
EarlyPD	Only EarlyPD speakers	Test early-stage-only training
EarlyPD+Private	EarlyPD plus external private data	Test benefit of additional external data

3.6 Speaker-Independent 5-Fold Split

The benchmark ensures proper evaluation through:

- A fixed, speaker-independent 5-fold split.
- Each validation and test set contains 6 EarlyPD and 6 HC speakers.
- 30 EarlyPD speakers are distributed across the test folds.
- Gender is balanced within each dataset and diagnostic group.
- HC and all-stage PD test sets are matched for comparison.

3.7 Models Evaluated

Three representative models spanning different paradigms are evaluated.

Table 6: Representative models evaluated by the benchmark.

Model	Type	Key Feature
BDHPD	Self-supervised learning	Pretrained SSL speech representation
InceptionPD	Vision-pretrained	Spectrogram images with transfer learning
RECA-PD	Explainable AI	Cross-attention with interpretability

3.8 Multi-Dimensional Evaluation

The benchmark introduces comprehensive evaluation breakdowns across:

1. **Dataset-level:** PC-GITA versus NeuroVoz.
2. **Aggregation-level:** 3 vowels, 3 sentences, and 10 sentences per speaker.
3. **Gender:** Female versus Male.
4. **Disease Stage:** EarlyPD versus All-stage PD.

4 Results and Key Findings

4.1 Main Results

Table 7: Main benchmark results.

Model	Best Task	F1 (Avg)	AUC (Avg)	Key Insight
RECA-PD	DDK, Sentence	0.69	0.74	Explainable and competitive
InceptionPD	Vowel	0.66	0.61	Good for spectrogram-based tasks
BDHPD	Variable	0.65	0.62	Underperforms in single-task setting

Key Findings:

1. **Training Data Composition Matters:** Comparison of EarlyPD and AllPD-sub showed that DDK and vowel tasks improved with early-stage-only training, whereas sentence reading benefited from broader disease-stage variability.
2. **Adding PD Speakers Improves Performance:** AllPD, which adds non-EarlyPD speakers, achieved the highest average F1 on DDK and sentence tasks. EarlyPD+Private improved AUC on shared tasks.
3. **Model Performance Varies by Task:** RECA-PD performs particularly well on DDK and sentence tasks, whereas InceptionPD shows competitive AUC on sustained vowels.

4.2 Multi-Dimensional Evaluation Results

4.2.1 Dataset-Level Performance Gap

Table 8: Dataset-level performance comparison.

Dataset	Best F1	Best AUC	Gap
PC-GITA	0.74 ± 0.04	0.83 ± 0.05	+0.09 F1
NeuroVoz	0.65 ± 0.03	0.68 ± 0.02	+0.15 AUC

This gap highlights **dataset generalization** as a key priority for future research.

4.2.2 Aggregation-Level Findings

The benchmark reports several important observations regarding aggregation:

- Aggregating multiple recordings mitigates intra-speaker variability.
- Larger gains are observed in AUC than in F1 when more samples are aggregated.
- BDHPD and InceptionPD improve with more aggregated samples.
- RECA-PD shows a small degradation with 3-sample aggregation but improves with 10 sentences.

4.2.3 Gender Analysis

The gender analysis produced the following observations:

- Female speakers outperform male speakers across all models and tasks.
- This reverses prior findings that reported higher male accuracy.
- Gender groups had similar H&Y stage, TAD, and MDS-UPDRS scores.
- The findings imply that dataset variability may affect fairness and therefore warrant further investigation.

4.2.4 Disease Stage Comparison

Table 9: Performance difference between all-stage and EarlyPD settings.

Model	DDK Δ	Vowel Δ	Sentence Δ
BDHPD	+0.04	+0.05	+0.07
InceptionPD	+0.01	-0.03	+0.05
RECA-PD	-0.02	+0.01	+0.08

The results show that:

- Most deltas are positive, indicating better performance for all-stage PD than EarlyPD.
- The sentence task shows the largest performance gap.
- Sentence detection benefits from broader disease-stage variability.
- EarlyPD detection is therefore more challenging than all-stage PD detection.

5 Broader Context and Related Work

5.1 Current State of the Field

A systematic umbrella review of 22 reviews covering 458 primary studies found:

- **Most common ML algorithms:** SVM, ANN, and KNN.
- **Most common DL algorithms:** CNN and DNN.
- **Performance ranges:** AUC from 0.5–1.0 and F1 from 0.690–0.997.
- **Key weakness:** Methodological quality is generally low, with insufficient reporting of critical clinical factors.

5.2 Transition in the Field

The field is transitioning from handcrafted feature-based pipelines toward self-supervised and representation-learning frameworks.

Table 10: Evolution of machine learning approaches for PD speech detection.

Era	Approach	Strengths	Limitations
Classical ML	Handcrafted features + SVM/RF	Interpretability and efficiency on small datasets	Fragile to recording conditions
Deep Learning	CNNs, RNNs on spectrograms	Automatic feature extraction	Black-box behavior and data requirements
Foundation Models	Wav2Vec 2.0, HuBERT, Whisper	Generalization and cross-lingual capabilities	Limited interpretability and high computational cost

5.3 Persistent Challenges

Several challenges remain persistent in the field:

1. **Dataset Heterogeneity:** Different languages, recording conditions, and demographic characteristics.
2. **Class Imbalance:** Many datasets are skewed toward male speakers or specific H&Y stages.
3. **Limited External Validation:** Most studies rely on internal validation.
4. **Lack of Interpretability:** Black-box models can limit clinician trust.
5. **Inconsistent Reporting:** Confidence intervals and variance measures are absent from many studies.

6 Critical Analysis of the Benchmark

6.1 Strengths

The benchmark has several important strengths:

1. **First of its Kind:** Specifically designed for EarlyPD detection.

2. **Clinically Grounded:** Uses validated EarlyPD criteria of H&Y ≤ 2 and TAD ≤ 5 years.
3. **Transparent and Replicable:** Uses public resources and fixed speaker-independent splits.
4. **Multi-dimensional Evaluation:** Considers dataset, aggregation, gender, and disease stage.
5. **Multiple Training Settings:** Isolates the effects of training data composition.
6. **Open + Private Tracks:** Balances accessibility with real-world constraints.

6.2 Limitations

1. **Limited Datasets:** Only PC-GITA and NeuroVoz meet the benchmark criteria.
2. **Single-Task Training:** Future work should explore multi-task strategies.
3. **No Spontaneous Speech:** Spontaneous speech is not included because of limited support in existing methods.
4. **Small EarlyPD Sample:** Only 30 speakers are distributed across the test folds.
5. **Language Bias:** Both open-track datasets are Spanish.
6. **Limited Geographic Diversity:** The available datasets are primarily European.

6.3 Significance

The benchmark addresses the "central, recognized, and long-standing limitation in speech-based PD detection research"—the lack of principled comparability.

By establishing standardized evaluation protocols, it enables:

- Fair cross-method comparisons.
- Reproducible research.
- Clinically meaningful evaluation through aggregation-level and gender-stratified analysis.
- Identification of dataset generalization challenges.

7 Future Directions

7.1 Immediate Priorities

1. Adoption of the benchmark as a shared reference point.
2. Multi-dimensional evaluation reporting in all future work.
3. Expansion to more languages and datasets.
4. Inclusion of spontaneous speech tasks.

7.2 Long-Term Needs

1. Large, multilingual, openly accessible datasets.
2. Standardized validation protocols, particularly external and subject-independent validation.
3. Fairness- and diversity-aware modeling with subgroup analysis.
4. Interpretable AI frameworks to improve clinician trust.
5. Prospective clinical validation and appropriate regulatory pathways.

8 Conclusion

The benchmark by Zhong et al. represents a significant advance in speech-based Parkinson's disease detection research. By establishing the first standardized evaluation framework specifically for EarlyPD, it addresses the fragmentation that has long hindered meaningful comparison across studies.

The results confirm that EarlyPD detection is more challenging than all-stage PD detection and highlight dataset generalization as a key priority. The strong performance of the explainable RECA-PD model demonstrates that explainable AI design need not necessarily compromise predictive performance, which is an important consideration for eventual clinical adoption.

Future progress will depend on community-wide adoption of standardized benchmarks and continued efforts to develop large, multilingual, accessible datasets with consistent evaluation protocols.

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